

# Frontend Roadmap

## HTML

- HTML Structure & Syntax
- Tags, Attributes (id, class, href, src)
- Text Formatting (<p>, <h1>-<h6>, <em>, <strong>)
- Head Tags ( <title>, <meta>, <link>, <style>, <script> )
- Lists: Ordered, Unordered, Description
- Tables: Basic and Advanced (colspan, rowspan)
- Hyperlink Tag ( <a> )
- Form Tags ( <form>, <input>, <textarea>, <button> )
- Media Tags ( <img>, <video>, <audio>, <source> )
- Semantic ( <nav>, <main> ), Non-Semantic Tags ( <div>, <span> )

## CSS

- Intro to CSS: Syntax ( Declaration , Rule , Property , Value )
- Connection Methods : Inline, Internal, External
- CSS Selectors: Tag, Class, ID, Grouping
- Pseudo-classes & Elements ( :hover, ::before )
- CSS Properties: color, bg-color, font, text
- Box Model: margin, padding, border, content
- Overflow, Box-Sizing, Max/Min Width & Height
- Positioning: static, relative, absolute, fixed, sticky
- Display Types: block, inline, inline-block, none
- Display Flex : Justify Content, Align Items, Direction , Wrap
- Display Grid: Structure, Template Columns, Rows , Areas
- CSS Transitions , Transform and Animations
- Media Queries & Responsive Design
- CSS Variables , Functions

## **BOOTSTRAP**

- Bootstrap Setup
- Responsive Layouts Grid System
- Typography, Spacing, Colors, Buttons
- Forms, Input Groups, Utilities
- Components: Navbar, Card, Carousel, Modal

## **JAVASCRIPT**

- Introduction to JavaScript: Setup, Console, Syntax
- Variables (var, let, const) , Scope , Hoisting and Data Types
- Type Coercion, Template Literals
- Operators: Arithmetic,.... Logical ,.... Ternary
- Conditional Statements (if, else, nested, switch, ternary)
- Loops Statements : for, while, do while
- Control Statements (break, continue)
- Functions: Declaration, Expression, Parameters, Arguments
- Return, Function Types ( arrow , callback , higher order )
- Arrays: Basics, Indexing, Looping
- Array Methods: push, pop, shift, unshift, slice, splice
- Array Methods Advanced: map, filter, reduce
- Objects: Properties, Methods, Nested Objects
- Spread, Rest, Destructuring Arrays
- Object Destructuring, Optional Chaining
- Sets & Maps: Usage , Methods
- JSON: Structure, Parse, Stringify
- Error Handling: try, catch, finally
- DOM & BOM introduction
- DOM Basics: Selecting, Modifying Elements , Style , ClassList
- DOM Events: Listener ,Event Object , Delegation , Bubbling
- Timing Functions: setTimeout, setInterval , Promise , Async
- ES6 Modules: import, export
- Local Storage & Session Storage

## **REACT**

- React Setup (Vite), Folder Structure
- JSX Syntax and Rules
- Components: Functional, Class
- Hooks: useState, useEffect
- Props and Props Drilling
- Conditional Rendering (if, ? :, &&)
- Rendering Lists with map() and Keys
- Forms in React, Controlled Components
- Routing with react-router-dom

## **Others**

- Mini Project: HTML + CSS Portfolio Site
- Git, GitHub Profile, Deployment (Netlify/Vercel)
- Resume Building & LinkedIn Optimization
- Mock Interviews (Frontend Tech Round)
- JS & React Interview Questions Practice
- HR Interview Questions & Communication Tips
- DSA for Frontend (Array, String, Object-based)
- Final Presentation, Review & Certificate Process